BLUESTOCK MUTUAL FUND ANALYTICS PLATFORM

End-to-End Data Analytics & Business Intelligence Capstone Project

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Submitted By

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Project Duration

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Submitted To

Bluestock Fintech

Data Analytics Internship Program

Technologies Used

- Python
- Pandas
- NumPy
- SQL
- Power BI
- Jupyter Notebook
- Git
- GitHub
- Microsoft Excel

Project Domain

Financial Analytics | Business Intelligence | Mutual Fund Analytics

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Version 1.0

June 2026

Certificate of Completion

This report presents the work completed as part of the **Bluestock Mutual Fund Analytics Platform Capstone Project** during the Data Analyst Internship at Bluestock Fintech.

I hereby declare that the work presented in this report is my original work and has been completed using publicly available mutual fund datasets along with data engineering, analytics, and visualization techniques learned during the internship. The project has been developed independently for academic and professional learning purposes.

The analysis, dashboard development, advanced financial metrics, business insights, and recommendations included in this report were implemented using Python, SQL, and Microsoft Power BI.

This report has not been submitted previously for any other certification, degree, or internship evaluation.

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Executive Summary

The **Bluestock Mutual Fund Analytics Platform** is an end-to-end business intelligence solution developed to analyze the Indian mutual fund industry using historical fund performance, investor transactions, portfolio holdings, Assets Under Management (AUM), Systematic Investment Plan (SIP) trends, and benchmark performance. The primary objective of the project is to transform raw financial datasets into meaningful business insights that support investment analysis and strategic decision-making.

The project integrates information from twelve different datasets covering fund metadata, daily NAV history, investor transactions, benchmark indices, SIP inflows, category-wise investments, portfolio holdings, performance metrics, and risk indicators. A complete ETL pipeline was implemented in Python to clean, validate, transform, and prepare the datasets for analytical processing.

Interactive dashboards were developed in Microsoft Power BI to provide comprehensive visibility into industry growth, fund performance, investor behavior, and SIP trends. Four dashboard pages were designed to allow users to explore key financial metrics through dynamic filters and visualizations.

To enhance analytical capabilities, advanced financial techniques including Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling 90-Day Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration using the Herfindahl-Hirschman Index (HHI), and a Mutual Fund Recommendation System were implemented.

The final platform demonstrates the complete lifecycle of a professional data analytics project, from raw data ingestion and transformation to advanced analytics, interactive visualization, and business reporting. The solution provides actionable insights for investors, analysts, and financial institutions by enabling performance comparison, risk assessment, investor segmentation, and informed investment decision-making.

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Project Introduction

1. Introduction

The Indian mutual fund industry has experienced remarkable growth over the past decade, driven by increasing financial awareness, digital investment platforms, and the widespread adoption of Systematic Investment Plans (SIPs). With thousands of investment schemes offered by multiple Asset Management Companies (AMCs), investors often face challenges in comparing fund performance, evaluating investment risk, and making informed financial decisions.

Mutual fund companies generate vast amounts of data every day, including Net Asset Value (NAV) history, Assets Under Management (AUM), investor transactions, benchmark performance, portfolio holdings, and market trends. Analyzing this data manually is time-consuming and inefficient, making Business Intelligence (BI) and Data Analytics essential for transforming raw financial data into actionable insights.

The **Bluestock Mutual Fund Analytics Platform** was developed as an end-to-end data analytics solution to address these challenges. The project integrates multiple datasets covering mutual fund performance, investor behavior, SIP trends, benchmark indices, and portfolio composition into a unified analytics platform. Using Python for data processing and Microsoft Power BI for visualization, the platform enables users to explore industry trends, compare fund performance, evaluate portfolio risk, and understand investor behavior through interactive dashboards.

In addition to traditional business intelligence reporting, the project incorporates advanced financial analytics techniques such as Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration Analysis using the Herfindahl-Hirschman Index (HHI), and a rule-based Mutual Fund Recommendation System. These analytical models provide deeper insights into fund performance, investment risk, and investor engagement.

Overall, this project demonstrates the complete lifecycle of a modern data analytics solution, including data collection, ETL pipeline development, exploratory data analysis, advanced statistical modeling, dashboard development, and business reporting. The resulting platform supports data-driven decision-making for investors, analysts, and financial institutions while showcasing the practical application of data analytics in the financial services domain.

Business Problem & Project Objectives

2. Business Problem

The mutual fund industry manages large volumes of financial and investor data generated from daily Net Asset Value (NAV) updates, Assets Under Management (AUM), Systematic Investment Plan (SIP) transactions, benchmark indices, portfolio holdings, and investor activities. Although this data contains valuable insights, it is often distributed across multiple sources and is difficult to analyze without an integrated analytics platform.

Investors and financial analysts face several challenges while evaluating mutual funds:

- Comparing the performance of multiple mutual fund schemes across different categories.
- Assessing the risk associated with investments using financial metrics rather than relying only on returns.
- Understanding investor behavior and transaction patterns across different demographic segments.
- Monitoring industry growth through AUM, SIP inflows, and benchmark market performance.
- Identifying high-performing funds based on multiple performance indicators instead of a single return metric.
- Generating business insights quickly through interactive dashboards instead of manual spreadsheet analysis.

Without a centralized analytics solution, decision-making becomes slower, less accurate, and highly dependent on manual reporting.

3. Project Objectives

The primary objective of this project is to develop an end-to-end Mutual Fund Analytics Platform that transforms raw financial datasets into meaningful business insights using data engineering, business intelligence, and advanced analytics techniques.

The specific objectives of the project are:

- Design a complete ETL pipeline to clean, transform, and integrate multiple mutual fund datasets.
- Develop an interactive Power BI dashboard to visualize industry trends, fund performance, investor analytics, and SIP growth.
- Analyze fund performance using financial metrics such as CAGR, Sharpe Ratio, Alpha, Beta, Standard Deviation, and Maximum Drawdown.
- Evaluate investment risk using advanced analytics techniques including Historical Value at Risk (VaR) and Conditional Value at Risk (CVaR).

- Perform Rolling 90-Day Sharpe Ratio analysis to measure risk-adjusted returns over time.
- Conduct Investor Cohort Analysis to study investment behavior across different investor groups.
- Analyze SIP continuity to identify investors at risk of discontinuing their investments.
- Measure portfolio diversification using the Herfindahl-Hirschman Index (HHI).
- Develop a rule-based Mutual Fund Recommendation System based on investor risk appetite.
- Generate actionable business insights that support investment analysis and data-driven decision-making.

By achieving these objectives, the project demonstrates the practical application of Data Analytics, Business Intelligence, and Financial Analytics in solving real-world investment analysis problems.

Dataset Description

The Bluestock Mutual Fund Analytics Platform integrates multiple datasets representing different aspects of the Indian mutual fund ecosystem. These datasets were cleaned, transformed, and integrated through a structured ETL pipeline to support business intelligence, financial analytics, and interactive dashboard development.

Dataset Summary

Dataset	Description	Records
Fund Master	Master information for all mutual fund schemes including fund house, category, benchmark, expense ratio, risk category, and fund manager.	40
NAV History	Historical daily Net Asset Value (NAV) for each mutual fund scheme used to calculate returns and performance metrics.	46,000
AUM by Fund House	Assets Under Management (AUM) for major Asset Management Companies (AMCs).	48
Monthly SIP Inflows	Monthly Systematic Investment Plan (SIP) inflows, active SIP accounts, and SIP AUM.	48
Category Inflows	Monthly net inflows across different mutual fund categories.	288
Industry Folio Count	Industry-wide folio count and investor participation statistics.	48
Scheme Performance	Performance metrics including CAGR, Sharpe Ratio, Alpha, Beta, Standard Deviation, Maximum Drawdown, and AUM.	40
Investor Transactions	Individual investor transactions containing SIP, Lumpsum, and Redemption records along with demographic information.	32,778
Portfolio Holdings	Equity holdings for each mutual fund scheme including sector allocation and portfolio weights.	322
Benchmark Indices	Historical benchmark index values (NIFTY 50, NIFTY 100, NIFTY 500, BSE SmallCap, CRISIL Indices, etc.) used for fund comparison.	8,050+
Alpha-Beta Report	Calculated Alpha, Beta, and R-Squared values for all mutual fund schemes.	40
Fund Scorecard	Composite scorecard ranking schemes based on CAGR, Sharpe Ratio, Alpha, Expense Ratio, and Maximum Drawdown.	40

Data Characteristics

The combined dataset represents multiple dimensions of the mutual fund industry, including fund performance, market benchmarks, investor transactions, portfolio composition, and industry growth indicators. Historical NAV data spans multiple years, enabling return calculations, rolling performance analysis, and risk assessment.

Investor transaction records include demographic attributes such as state, city, city tier, age group, gender, payment mode, and KYC status, allowing behavioral and cohort-based analysis. Portfolio holdings provide sector-wise allocations and security weights, supporting portfolio concentration analysis through the Herfindahl-Hirschman Index (HHI).

The integration of these datasets enabled the development of a comprehensive analytics platform capable of generating business insights across operational, financial, and investment dimensions.

ETL Pipeline & Data Architecture

ETL Pipeline and Data Architecture

The Bluestock Mutual Fund Analytics Platform follows a structured **Extract, Transform, and Load (ETL)** pipeline to convert raw financial datasets into a clean, standardized, and analysis-ready data model. The pipeline was developed using Python and Pandas, ensuring data quality, consistency, and seamless integration across multiple datasets.

ETL Workflow

The complete data pipeline consists of the following stages:

1. Data Extraction

Raw CSV files containing mutual fund information, NAV history, investor transactions, portfolio holdings, benchmark indices, SIP inflows, AUM, and industry statistics were collected and imported into Python using the Pandas library.

2. Data Cleaning

The extracted datasets were cleaned through several preprocessing steps, including:

- Handling missing and null values
- Removing duplicate records
- Standardizing date formats
- Correcting inconsistent column names
- Converting data types
- Validating numeric fields
- Removing invalid or corrupted records

These steps ensured consistency across all datasets before analysis.

3. Data Transformation

After cleaning, multiple business transformations were performed, including:

- Daily return calculation from historical NAV values
- CAGR and return calculations
- Risk metric generation (Sharpe Ratio, Alpha, Beta, Standard Deviation, Maximum Drawdown)
- Benchmark return computation
- Feature engineering for investor analytics
- Fund scorecard generation
- Alpha-Beta report creation

These transformations enabled advanced financial analysis and dashboard reporting.

4. Data Integration

The processed datasets were integrated using common business keys such as:

- **AMFI Code** (Fund Identifier)
- **Scheme Name**
- **Transaction Date**
- **Benchmark Name**
- **Fund House**

This integration created a unified analytical model suitable for Power BI dashboards and advanced analytics.

5. Data Loading

The cleaned and transformed datasets were exported as processed CSV files and connected directly to Microsoft Power BI. Relationships between the datasets were established using **AMFI Code** and **Date**, enabling cross-filtering, drill-through functionality, and interactive business reporting.

Data Architecture

The overall project architecture follows the workflow below:

Raw CSV Files

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Data Extraction (Python & Pandas)

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Data Cleaning & Validation

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Data Transformation & Feature Engineering

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Processed Analytical Datasets

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Power BI Data Model

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Interactive Dashboards

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Advanced Analytics & Business Insights

The modular ETL architecture ensures scalability, maintainability, and reproducibility of the analytics workflow. By separating extraction, transformation, visualization, and advanced analytics into distinct stages, the platform supports efficient data processing while enabling future integration with databases, cloud platforms, or automated data pipelines.

Data Cleaning & Preprocessing

Data Cleaning & Preprocessing

Data quality plays a critical role in financial analytics, as inaccurate or inconsistent data can lead to misleading investment insights and incorrect business decisions. Before performing any analysis, all datasets underwent a comprehensive data cleaning and preprocessing process using Python and the Pandas library.

The objective of this stage was to convert raw financial data into a clean, consistent, and analysis-ready format suitable for Business Intelligence reporting and advanced financial analytics.

Data Cleaning Activities

The following preprocessing operations were performed across all datasets:

- Removed duplicate records to eliminate redundant observations.
- Identified and handled missing (NULL) values using appropriate strategies such as removal, imputation, or validation.
- Converted date columns into a standardized **YYYY-MM-DD** format for consistent time-series analysis.
- Corrected inconsistent column names by applying a uniform naming convention across all datasets.
- Standardized categorical values such as fund categories, transaction types, risk grades, and investment plans.
- Converted numerical columns including NAV, AUM, expense ratio, transaction amount, and benchmark values into appropriate numeric data types.
- Removed invalid or corrupted records that could negatively impact analytical results.
- Verified the uniqueness of primary identifiers such as **AMFI Code** and validated referential integrity across related datasets.
- Ensured consistency between benchmark data, fund master records, transaction history, and NAV datasets before integration.

Feature Engineering

In addition to cleaning, several analytical features were generated to support business intelligence and financial analysis.

These include:

- Daily percentage returns calculated from historical NAV values.
- Compound Annual Growth Rate (CAGR).
- Risk-adjusted performance metrics including Sharpe Ratio, Alpha, Beta, Standard Deviation, and Maximum Drawdown.
- Fund Scorecard for overall scheme ranking.
- Alpha-Beta report for benchmark comparison.

- Investor segmentation based on age group, gender, state, city tier, and transaction behavior.
- Monthly investment summaries for SIP and redemption analysis.
- Portfolio concentration metrics using stock holding weights.

Data Validation

After preprocessing, multiple validation checks were performed to ensure data reliability.

The validation process included:

- Verifying record counts after each transformation stage.
- Checking for remaining missing values.
- Confirming correct data types for all attributes.
- Validating relationships between datasets using **AMFI Code** and **Date**.
- Comparing calculated financial metrics with expected values to ensure computational accuracy.

The successful completion of the preprocessing stage produced a high-quality analytical dataset that served as the foundation for dashboard development, performance analytics, investor behavior analysis, and advanced risk modeling throughout the project.

Exploratory Data Analysis (EDA)

Exploratory Data Analysis

Exploratory Data Analysis (EDA) was conducted to understand the structure, distribution, and relationships within the mutual fund datasets before developing dashboards and advanced analytical models. The objective of this stage was to identify trends, detect anomalies, validate data quality, and discover meaningful business patterns.

The analysis was performed using Python libraries including **Pandas**, **NumPy**, **Matplotlib**, and **Seaborn**, enabling both statistical summaries and visual exploration of the data.

Industry Overview

The analysis revealed significant growth in the Indian mutual fund industry during the study period. Assets Under Management (AUM) and Systematic Investment Plan (SIP) inflows demonstrated a consistent upward trend, indicating increasing investor participation and confidence in mutual fund investments.

Fund houses managing diversified portfolios accounted for the largest share of industry AUM, while equity-oriented schemes attracted the highest long-term investments.

Fund Performance Analysis

Performance metrics showed considerable variation across different mutual fund categories.

Key observations include:

- Small-cap and mid-cap funds generated higher long-term returns but exhibited greater volatility.
- Large-cap funds provided comparatively stable returns with lower investment risk.
- Liquid and debt funds demonstrated lower volatility and consistent performance, making them suitable for conservative investors.
- Risk-adjusted performance varied significantly across schemes, emphasizing the importance of evaluating Sharpe Ratio, Alpha, and Maximum Drawdown rather than relying solely on returns.

Investor Behaviour Analysis

Analysis of investor transactions highlighted several behavioural trends:

- SIP investments constituted the largest proportion of total transactions, indicating a preference for disciplined long-term investing.
- Most investment activity originated from Tier-1 cities and financially active states.
- Investors in the 26–45 age group contributed the highest average investment amounts.
- Monthly transaction volumes showed stable investment patterns with periodic fluctuations corresponding to market conditions.

Market and SIP Trends

The comparison between monthly SIP inflows and benchmark market indices revealed a positive relationship between market performance and investor participation.

Category-wise inflow analysis indicated stronger investment preference toward equity-oriented mutual funds during periods of market growth, while debt and liquid funds gained popularity during periods of increased market uncertainty.

Key Findings from EDA

The exploratory analysis provided several important insights that guided subsequent dashboard development and advanced analytics:

- The mutual fund industry has experienced sustained growth in AUM and SIP participation.
- Equity mutual funds dominate long-term wealth creation despite higher market risk.
- Investor demographics significantly influence investment behaviour.
- Risk-adjusted performance metrics provide better fund evaluation than absolute returns alone.
- Historical trends reveal consistent growth in systematic investing, highlighting the increasing popularity of SIPs among retail investors.

The findings obtained during this stage established the analytical foundation for the Power BI dashboards and advanced financial models developed in the later phases of the project.

Dashboard Analysis – Industry Overview

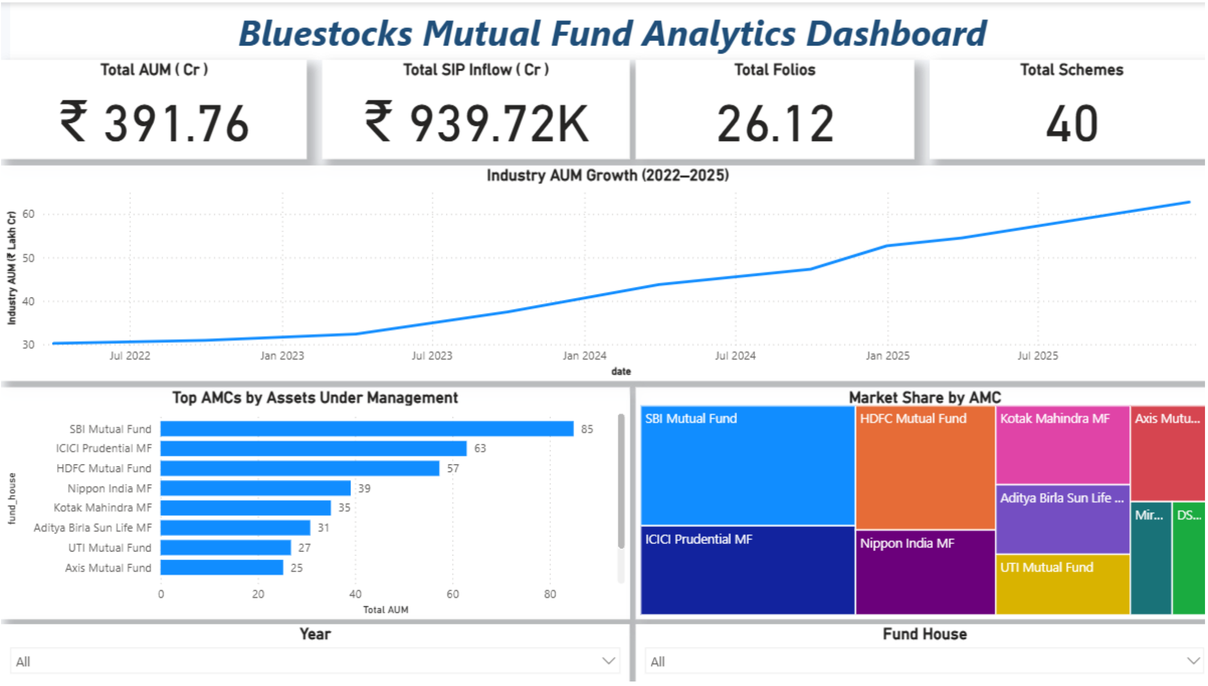

Fig 1 : Industry Overview

Dashboard Page 1: Industry Overview

The **Industry Overview Dashboard** provides a high-level summary of the Indian mutual fund industry's growth and performance. It is designed to help investors, analysts, and business stakeholders monitor key industry indicators through interactive visualizations and KPI cards.

The dashboard consolidates multiple datasets, including Assets Under Management (AUM), Systematic Investment Plan (SIP) inflows, industry folio counts, benchmark indices, and fund house statistics, enabling users to evaluate the overall health of the mutual fund market.

Key Performance Indicators (KPIs)

The dashboard displays four primary business KPIs:

- **Total Assets Under Management (AUM):** Represents the total value of assets managed by mutual funds, indicating the overall size of the industry.
- **Total SIP Inflows:** Measures monthly investments through Systematic Investment Plans, reflecting investor confidence and participation.
- **Industry Folio Count:** Shows the total number of investor accounts, providing insight into market penetration and retail participation.
- **Number of Mutual Fund Schemes:** Indicates the diversity of investment products available in the market.

These KPIs provide an instant overview of the industry's current performance.

Visualizations Included

The Industry Overview dashboard contains the following analytical visualizations:

- **Industry AUM Growth (2022–2025):** A line chart illustrating the growth trend of Assets Under Management over time.
- **Top Asset Management Companies (AMCs):** A horizontal bar chart comparing leading fund houses based on their total AUM.
- **Market Share Distribution:** A treemap showing the relative contribution of each AMC to the industry's total assets.
- **Interactive Filters:** Slicers for year and fund house allow users to perform focused analysis and compare specific time periods or organizations.

Business Insights

The dashboard highlights several important industry trends:

- The Indian mutual fund industry has experienced steady growth in Assets Under Management throughout the analysis period.
- A small number of large Asset Management Companies account for a significant proportion of total industry assets.
- Consistent SIP inflows indicate increasing investor confidence and long-term participation in mutual fund investments.
- Interactive filtering enables users to analyze industry performance across different years and fund houses, supporting more informed decision-making.

Overall, the Industry Overview Dashboard serves as the executive summary page of the analytics platform, providing stakeholders with a concise yet comprehensive view of industry performance before exploring detailed fund-level and investor-level analyses.

Dashboard Analysis – Fund Performance

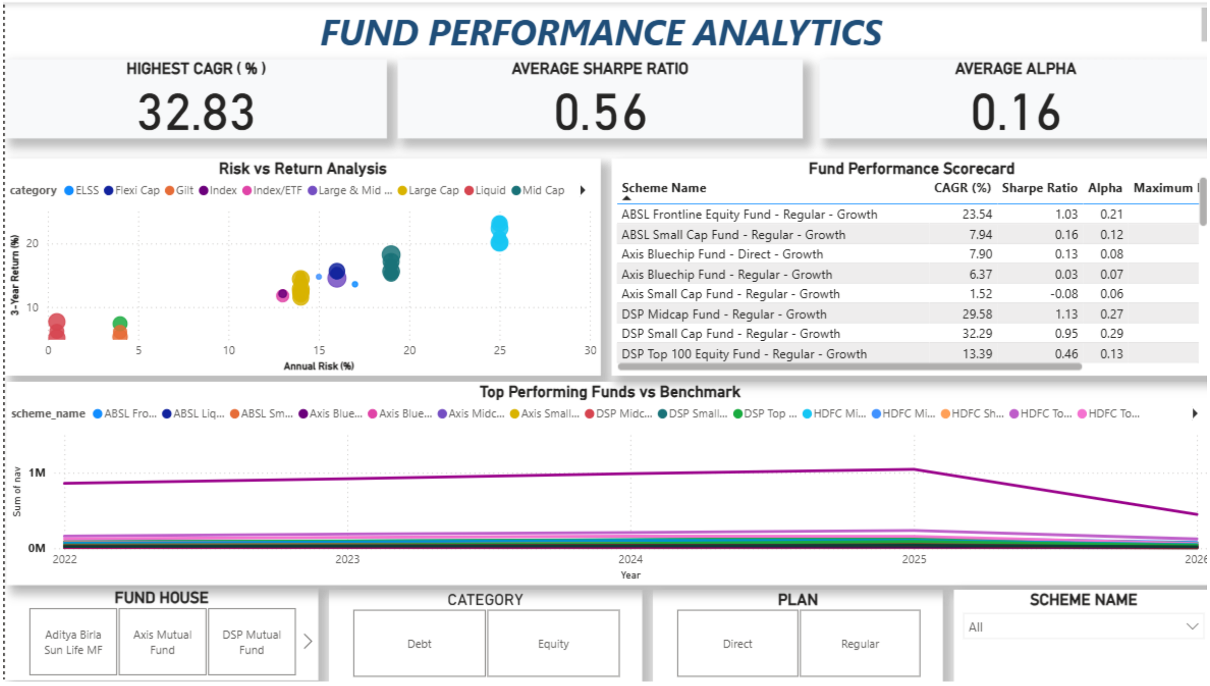

Fig 2 : Fund Performance

Dashboard Page 2: Fund Performance Analysis

The **Fund Performance Dashboard** provides a comprehensive evaluation of mutual fund schemes by combining return, risk, and risk-adjusted performance metrics. This dashboard enables investors and financial analysts to compare different funds using multiple performance indicators rather than relying solely on returns.

The dashboard integrates scheme performance data with calculated financial metrics such as CAGR, Sharpe Ratio, Alpha, Standard Deviation, Maximum Drawdown, and benchmark comparisons. Interactive filters allow users to analyze funds by fund house, category, investment plan, and scheme name.

Key Performance Indicators (KPIs)

The dashboard displays three important KPIs that summarize the overall performance of mutual fund schemes:

- **Highest CAGR (%):** Identifies the best-performing scheme based on long-term compounded annual growth.
- **Average Sharpe Ratio:** Represents the average risk-adjusted return across all schemes.
- **Average Alpha:** Measures the average excess return generated by fund managers compared to their respective benchmark indices.

These KPIs provide a quick summary of overall market performance and fund management efficiency.

Visualizations Included

The dashboard consists of the following analytical visualizations:

- **Risk vs Return Scatter Plot:** Compares Annual Risk (Standard Deviation) with 3-Year Returns while representing fund size through bubble size (AUM). This visualization helps identify funds delivering higher returns with relatively lower risk.
- **Fund Performance Scorecard:** A sortable comparison table displaying CAGR, Sharpe Ratio, Alpha, and Maximum Drawdown for each mutual fund scheme, enabling quick comparison across multiple performance metrics.
- **Top Performing Funds vs Benchmark:** A line chart comparing selected mutual fund NAV trends against benchmark index performance over time.
- **Interactive Filters:** Fund House, Category, Plan, and Scheme Name slicers allow users to perform customized analysis and compare specific funds.

Business Insights

Analysis of the dashboard reveals several important findings:

- Small-cap and mid-cap funds generally delivered higher long-term returns but exhibited greater volatility.
- Large-cap funds provided more stable performance with comparatively lower investment risk.
- Funds with higher Sharpe Ratios consistently generated superior risk-adjusted returns, making them more attractive for long-term investors.
- Positive Alpha values indicate fund managers who outperformed their benchmark indices, demonstrating effective portfolio management.
- Comparing NAV movements with benchmark indices enables investors to evaluate whether a fund consistently outperforms the broader market.

The Fund Performance Dashboard provides a balanced view of return, volatility, and risk-adjusted performance, allowing investors to make informed investment decisions based on multiple financial indicators rather than a single return metric.

Dashboard Analysis – Investor Analytics

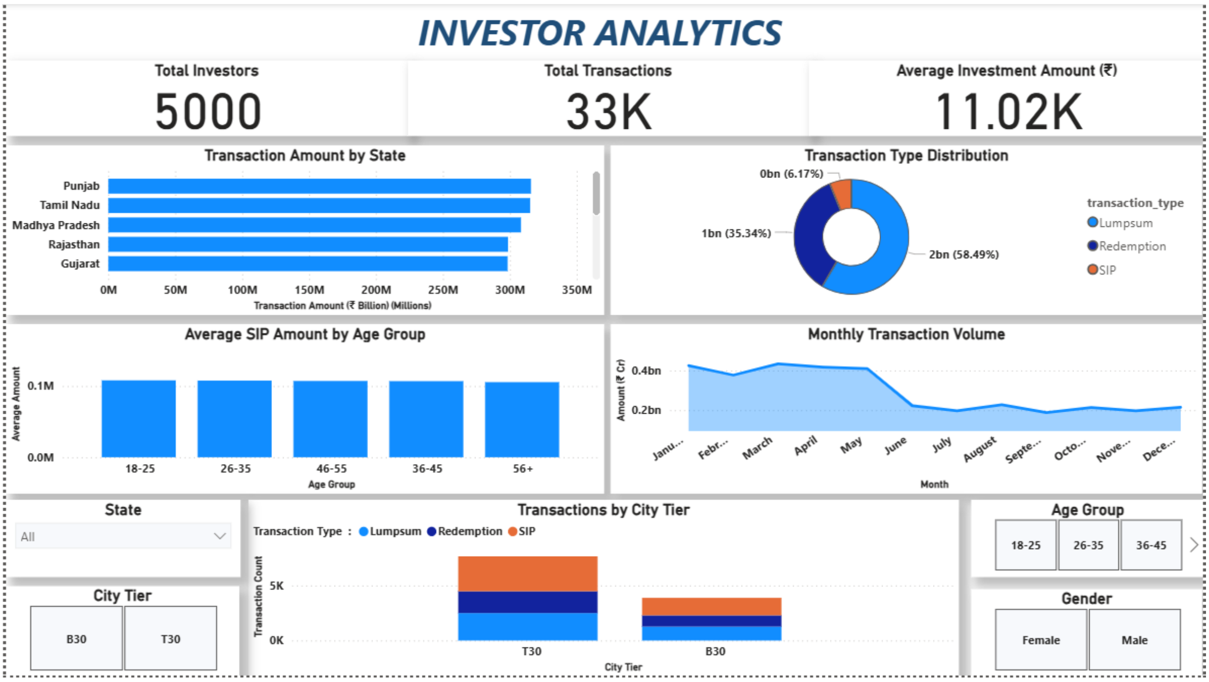

Fig 3 : Investor Analytics

Dashboard Page 3: Investor Analytics

The **Investor Analytics Dashboard** provides detailed insights into investor demographics, transaction behavior, and regional investment patterns. The objective of this dashboard is to understand how different groups of investors contribute to the mutual fund industry and identify trends that can support customer segmentation and business decision-making.

The dashboard combines investor transaction records with demographic information such as state, city tier, gender, age group, and transaction type. Interactive filters enable users to analyze investment behaviour across different customer segments.

Key Performance Indicator (KPI)

The dashboard displays the following primary KPI:

- **Average Investment Amount:** Represents the average amount invested per transaction, providing an overall view of investor participation and investment capacity.

Visualizations Included

The Investor Analytics Dashboard contains the following business visualizations:

- **Transaction Amount by State:** A horizontal bar chart showing the total investment amount across different states, helping identify regions with the highest investment activity.
- **Transaction Type Distribution:** A donut chart illustrating the proportion of SIP, Lumpsum, and Redemption transactions.

- **Average Investment Amount by Age Group:** A column chart comparing investment behaviour across different age groups.
- **Monthly Transaction Trend:** A line chart displaying transaction volume over time to identify seasonal or long-term investment patterns.
- **Transactions by City Tier:** A column chart comparing investment activity across Tier-1, Tier-2, and Tier-3 cities.
- **Interactive Filters:** State, Age Group, Gender, and City Tier slicers allow users to perform customized demographic analysis.

Business Insights

The dashboard provides several important insights into investor behaviour:

- Certain states contribute significantly higher investment volumes, indicating stronger market penetration and financial awareness.
- SIP transactions account for the majority of investment activity, demonstrating investors' preference for disciplined long-term wealth creation.
- Investors in the **26–45 years** age group contribute the highest average investment amounts, making them the primary target segment for mutual fund companies.
- Tier-1 cities continue to dominate investment activity; however, increasing participation from Tier-2 and Tier-3 cities highlights the growing adoption of digital investment platforms.
- Monthly transaction trends indicate consistent investor participation, reflecting stable confidence in mutual fund investments.

The Investor Analytics Dashboard enables financial institutions to better understand customer behaviour, identify high-value investor segments, and develop targeted investment strategies based on demographic and regional insights.

<u>**Dashboard Analysis – SIP & Market Trends**</u>

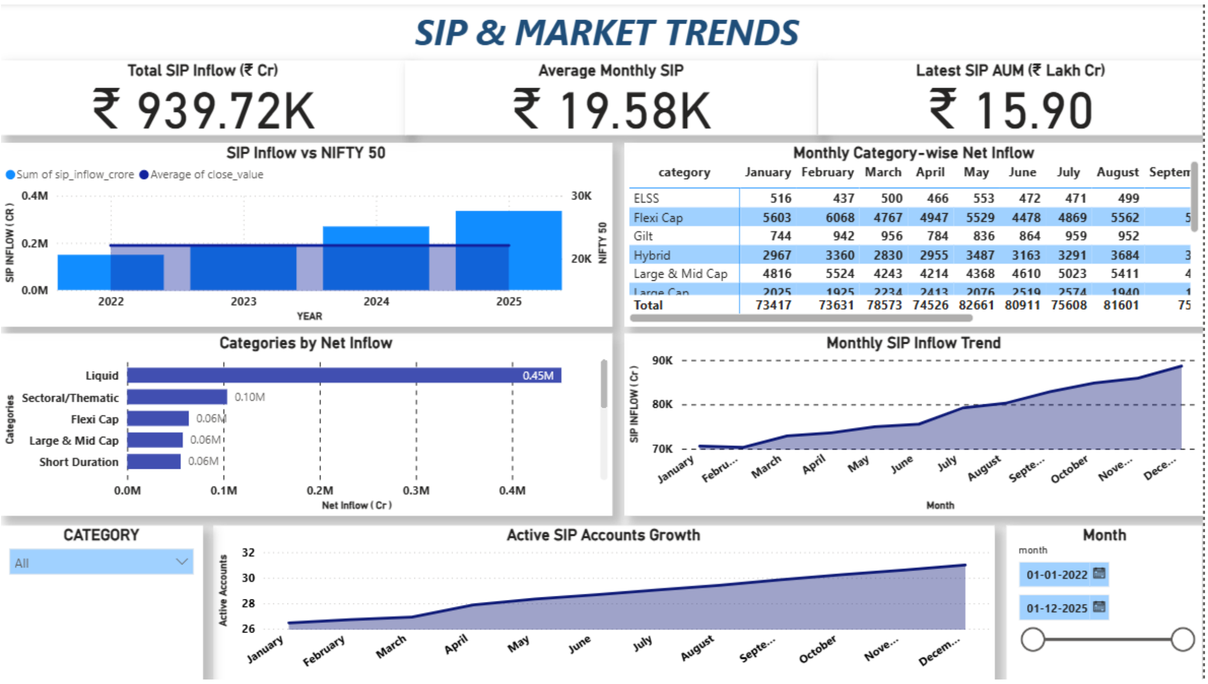

Fig 4 : SIP & Market Trends

Dashboard Page 4: SIP & Market Trends

The **SIP & Market Trends Dashboard** provides insights into the growth of Systematic Investment Plans (SIPs), industry-wide investment trends, category-wise fund inflows, and benchmark market performance. This dashboard helps investors and financial analysts understand how systematic investments have evolved over time and how market movements influence investment behaviour.

The dashboard combines SIP statistics, category-wise inflow data, benchmark indices, and market performance into a single analytical view, enabling users to evaluate long-term industry trends and investment patterns.

Key Performance Indicators (KPIs)

The dashboard displays the following key industry metrics:

- **Total SIP Inflow:** Represents the cumulative value of investments made through Systematic Investment Plans during the analysis period.
- **Total Category Inflow:** Displays the overall investment across different mutual fund categories, highlighting investor preferences.
- **Average Industry AUM:** Indicates the average Assets Under Management across the selected period, reflecting the growth of the mutual fund industry.

These KPIs provide a quick overview of industry expansion and investor participation.

Visualizations Included

The SIP & Market Trends Dashboard consists of the following visualizations:

- **SIP Inflow vs NIFTY 50 Trend:** A combined chart comparing monthly SIP inflows with NIFTY 50 index performance to understand the relationship between market movements and investor participation.
- **Category-wise Net Inflow:** A horizontal bar chart ranking mutual fund categories based on total net investments.
- **Monthly Category Inflow Heatmap:** A heatmap illustrating monthly investment patterns across different fund categories, enabling quick identification of seasonal and category-specific trends.
- **Monthly SIP Inflow Trend:** A line chart showing the growth of SIP investments over time.
- **Active SIP Accounts Growth:** A line chart tracking the increase in active SIP accounts throughout the analysis period.
- **Interactive Filter:** A category slicer allowing users to analyze individual mutual fund categories.

Business Insights

The dashboard highlights several significant industry trends:

- Monthly SIP inflows have shown a consistent upward trajectory, indicating increasing investor confidence and growing adoption of disciplined investment strategies.
- Equity-oriented categories attracted the highest net inflows, reflecting investors' preference for long-term capital appreciation.
- Market performance and SIP participation exhibit a positive relationship, with stronger market conditions generally accompanied by higher investment activity.
- The number of active SIP accounts has steadily increased over time, demonstrating the expanding retail investor base within the mutual fund industry.
- Category-wise analysis enables Asset Management Companies (AMCs) to identify investor preferences and design products that align with changing market demand.

The SIP & Market Trends Dashboard provides a comprehensive overview of investment behaviour, market performance, and industry growth, supporting strategic decision-making for investors, fund managers, and financial institutions.

Advanced Financial Analytics

Advanced Analytics – Risk Assessment and Performance Evaluation

Beyond traditional business intelligence reporting, the project incorporates advanced financial analytics techniques to evaluate investment risk, portfolio performance, and investor behaviour. These analytical models provide deeper insights that support informed investment decisions and demonstrate the practical application of quantitative finance concepts.

Historical Value at Risk (VaR)

Historical Value at Risk (VaR) was calculated at the **95% confidence level** for all 40 mutual fund schemes using historical daily returns.

VaR estimates the maximum expected loss that a fund may experience over a single trading day under normal market conditions with a 95% confidence level.

The calculation was performed using the **5th percentile** of the historical daily return distribution.

Formula

VaR (95%) = 5th Percentile of Historical Daily Returns

The analysis revealed that **Small Cap** and **Mid Cap** funds exhibited the highest downside risk due to their greater market volatility, whereas **Liquid Funds** recorded the lowest Value at Risk, reflecting their relatively stable performance.

Conditional Value at Risk (CVaR)

Conditional Value at Risk (CVaR), also known as **Expected Shortfall**, measures the average loss that occurs when returns fall below the VaR threshold.

Unlike VaR, which identifies only the loss threshold, CVaR evaluates the severity of losses during adverse market conditions.

Formula

CVaR = Mean(Returns < VaR)

The CVaR results indicated that equity-oriented small-cap funds experience larger losses during extreme market movements, while debt and liquid funds remain comparatively resilient.

Rolling 90-Day Sharpe Ratio

To evaluate risk-adjusted performance over time, a **Rolling 90-Day Sharpe Ratio** was calculated for selected high-performing mutual funds.

The rolling Sharpe Ratio measures how efficiently a fund generates excess returns relative to its volatility over a moving 90-day window.

Formula

Rolling Sharpe = (Rolling Mean Return ÷ Rolling Standard Deviation) × √252

This analysis provided a dynamic view of fund performance instead of relying on a single historical Sharpe Ratio value.

Key observations include:

- Large-cap funds demonstrated relatively stable Sharpe Ratios throughout the analysis period.
- Small-cap funds produced higher peaks but also experienced greater fluctuations, reflecting increased market sensitivity.
- Risk-adjusted performance improved significantly during sustained bull market phases.

Business Value

The implementation of advanced financial analytics extends the project beyond conventional reporting by enabling:

- Quantitative measurement of investment risk.
- Identification of funds vulnerable to market downturns.
- Continuous monitoring of risk-adjusted performance.
- Better comparison of mutual fund schemes using statistical metrics.
- Data-driven investment recommendations based on both return and risk.

These analytical models significantly enhance the decision-making capabilities of investors, financial analysts, and portfolio managers by providing a comprehensive understanding of fund performance under varying market conditions.

<u>**Advanced Analytics – Investor Behaviour and Portfolio Analysis**</u>

Investor Cohort Analysis, SIP Continuity, Portfolio Concentration & Fund Recommendation System

To extend the analytical capabilities of the Mutual Fund Analytics Platform, several advanced business analytics techniques were implemented to understand investor behaviour, evaluate portfolio diversification, and provide personalized investment recommendations.

Investor Cohort Analysis

Investor Cohort Analysis was performed by grouping investors according to the year of their first investment transaction. This technique helps identify long-term investment behaviour and compare how different investor groups contribute to mutual fund investments over time.

For each cohort, the following metrics were calculated:

- Average SIP Amount
- Total Investment Value
- Most Preferred Mutual Fund Scheme

The analysis revealed that investors who entered the market during earlier periods contributed significantly higher cumulative investments than recent cohorts. It also identified the most preferred mutual fund schemes among different investor groups, providing valuable insights for customer segmentation and marketing strategies.

SIP Continuity Analysis

A SIP Continuity Analysis was conducted to evaluate investor consistency in making periodic investments.

Investors with six or more SIP transactions were analyzed, and the average gap between consecutive SIP payments was calculated.

The classification criteria were:

- **Healthy Investor:** Average gap ≤ 35 days
- **At-Risk Investor:** Average gap > 35 days

This analysis enables financial institutions to proactively identify investors who may discontinue their SIPs and design retention campaigns or reminder systems to improve customer engagement.

Portfolio Concentration Analysis (HHI)

Portfolio diversification was evaluated using the **Herfindahl-Hirschman Index (HHI)**.

HHI measures portfolio concentration by summing the squares of individual stock weight percentages.

Formula

$HHI = \Sigma (Weight_i^2)$

The analysis classified equity mutual funds into different concentration categories:

- Diversified Portfolio
- Moderately Concentrated Portfolio
- Highly Concentrated Portfolio

Funds with higher HHI values were found to have larger allocations toward a limited number of securities, indicating increased concentration risk. Conversely, diversified portfolios exhibited lower concentration scores, reducing exposure to individual stock-specific risk.

Mutual Fund Recommendation System

A rule-based Mutual Fund Recommendation System was developed to recommend suitable investment options based on an investor's risk appetite.

Investors are categorized into three risk profiles:

- Low Risk
- Moderate Risk
- High Risk

The recommendation engine ranks funds using multiple performance indicators, including:

- Sharpe Ratio
- Three-Year Return
- Expense Ratio
- Risk Category

Based on these metrics, the system recommends the **Top 3 mutual fund schemes** for each investor profile.

This demonstrates how financial analytics can be integrated with simple decision-support logic to assist investors in selecting appropriate investment options.

Business Value

The advanced analytical models implemented in this project significantly enhance the overall business value of the platform by:

- Identifying investor behaviour across different cohorts.
- Detecting investors at risk of discontinuing SIP investments.
- Measuring portfolio diversification using quantitative techniques.
- Supporting personalized investment recommendations based on risk appetite.

- Enabling financial institutions to improve customer engagement and investment advisory services through data-driven insights.

Collectively, these advanced analytics transform the platform from a traditional reporting dashboard into a comprehensive financial decision-support system.

Key Business Insights

Key Business Insights

The analytical findings obtained from the Mutual Fund Analytics Platform provide valuable business insights into industry growth, fund performance, investor behaviour, portfolio risk, and investment trends. These insights can assist Asset Management Companies (AMCs), financial advisors, and investors in making informed, data-driven decisions.

1. Strong Growth in the Mutual Fund Industry

The analysis indicates a consistent increase in Assets Under Management (AUM), SIP inflows, and investor folio counts throughout the study period. This reflects growing investor confidence and the increasing adoption of mutual funds as a preferred long-term investment vehicle.

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2. Risk and Return Must Be Evaluated Together

Funds delivering the highest returns were not always the best investment options. Risk-adjusted metrics such as Sharpe Ratio, Alpha, and Maximum Drawdown revealed that several funds with moderate returns provided superior performance when risk was considered.

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3. Small-Cap Funds Offer Higher Returns but Greater Risk

Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), and Standard Deviation analysis demonstrated that Small-Cap and Mid-Cap funds generated higher long-term returns but were also significantly more volatile than Large-Cap and Liquid funds. Investors should therefore align fund selection with their individual risk appetite.

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4. SIP Remains the Preferred Investment Method

Investor transaction analysis showed that Systematic Investment Plans (SIPs) accounted for the majority of investment transactions. The continuous growth in active SIP accounts indicates that investors increasingly prefer disciplined and long-term wealth creation over one-time investments.

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5. Investor Demographics Influence Investment Behaviour

The majority of investment activity originated from the **26–45 years** age group and Tier-1 cities. However, growing participation from Tier-2 and Tier-3 cities suggests expanding financial inclusion and increasing adoption of digital investment platforms across India.

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6. Portfolio Diversification Reduces Investment Risk

Portfolio Concentration Analysis using the Herfindahl-Hirschman Index (HHI) revealed that highly concentrated portfolios are exposed to greater stock-specific risk. Diversified portfolios demonstrated better risk distribution, making them more resilient during periods of market volatility.

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7. Personalized Recommendations Improve Investment Decisions

The rule-based Mutual Fund Recommendation System successfully identified suitable investment options for different investor risk profiles by considering Sharpe Ratio, historical returns, expense ratio, and risk category. This demonstrates the practical application of analytics in personalized financial advisory services.

Business Impact

The insights generated through this platform enable financial institutions to:

- Improve investment decision-making using data-driven performance analysis.
- Identify high-performing funds while effectively managing investment risk.
- Better understand investor demographics and behavioural patterns.
- Monitor industry growth through AUM, SIP, and benchmark performance.
- Develop targeted customer engagement and retention strategies.
- Support personalized investment recommendations based on investor risk profiles.

Overall, the Bluestock Mutual Fund Analytics Platform transforms complex financial datasets into meaningful business intelligence, enabling investors, analysts, and financial institutions to make more informed, accurate, and strategic investment decisions.

Limitations & Future Scope

Project Limitations and Future Scope

Project Limitations

Although the Bluestock Mutual Fund Analytics Platform successfully demonstrates end-to-end financial analytics and business intelligence, the current implementation has certain limitations that provide opportunities for future enhancement.

1. Static Dataset

The project is based on historical datasets collected for analytical purposes. The dashboards do not update automatically with live market data, requiring manual data refresh for new information.

2. Limited Historical Coverage

The analysis is restricted to the available historical period. A longer time horizon would enable more comprehensive market cycle analysis and improve the accuracy of long-term financial models.

3. Rule-Based Recommendation System

The mutual fund recommendation engine uses predefined business rules based on risk category, Sharpe Ratio, returns, and expense ratio. It does not currently employ Machine Learning techniques to generate personalized investment recommendations.

4. Simplified Risk Modeling

The project includes advanced financial metrics such as Historical VaR, CVaR, and Rolling Sharpe Ratio. However, more sophisticated quantitative models such as Monte Carlo Simulation, Black-Litterman Portfolio Optimization, or GARCH volatility forecasting were beyond the scope of this project.

5. Limited Portfolio Optimization

Portfolio analysis focuses on concentration measurement using the Herfindahl-Hirschman Index (HHI). The project does not perform asset allocation optimization or portfolio rebalancing recommendations.

Future Scope

The Mutual Fund Analytics Platform can be further enhanced by incorporating modern data engineering, machine learning, and cloud technologies.

Potential future improvements include:

- Integration with live financial APIs such as AMFI, NSE, BSE, or Yahoo Finance for real-time data updates.
- Development of an AI-powered recommendation system using Machine Learning algorithms and investor profiling.
- Implementation of predictive analytics to forecast fund performance, NAV trends, and SIP growth.
- Portfolio optimization using Modern Portfolio Theory (MPT) and Efficient Frontier analysis.

- Deployment of the platform on cloud services such as Microsoft Azure, AWS, or Google Cloud for automated data pipelines and scalable reporting.
- Development of a web-based analytics portal using Streamlit or Flask to provide interactive access for investors and financial advisors.
- Automated report generation and scheduled dashboard refreshes using Power BI Service and cloud-based ETL pipelines.
- Integration of Natural Language Query (NLQ) capabilities to allow users to interact with dashboards using conversational questions.

Overall Project Outcome

The Bluestock Mutual Fund Analytics Platform successfully demonstrates the practical application of Data Engineering, Business Intelligence, Financial Analytics, and Data Visualization in solving real-world investment analysis problems.

The project combines structured ETL pipelines, advanced financial risk metrics, interactive Power BI dashboards, and business-oriented analytics into a single end-to-end solution. It provides investors, analysts, and financial institutions with actionable insights that support informed investment decisions while serving as a scalable foundation for future enhancements using Artificial Intelligence and cloud technologies.

Conclusion

The **Bluestock Mutual Fund Analytics Platform** successfully demonstrates the complete lifecycle of a modern data analytics project by integrating data engineering, financial analytics, business intelligence, and interactive visualization into a unified solution.

The project began with the collection and integration of twelve independent datasets containing mutual fund information, historical NAV data, investor transactions, portfolio holdings, benchmark indices, SIP statistics, AUM data, and fund performance metrics. Through a structured ETL pipeline developed in Python, the raw datasets were cleaned, transformed, validated, and converted into analysis-ready data models.

Comprehensive exploratory data analysis provided valuable insights into industry growth, investor behaviour, fund performance, and market trends. Interactive dashboards developed in Microsoft Power BI enabled users to monitor industry KPIs, compare mutual fund performance, analyze investor demographics, evaluate SIP trends, and explore category-wise investment patterns through intuitive visualizations and dynamic filters.

To extend the analytical capabilities beyond traditional reporting, advanced financial models including Historical Value at Risk (VaR), Conditional Value at Risk (CVaR), Rolling 90-Day Sharpe Ratio, Investor Cohort Analysis, SIP Continuity Analysis, Portfolio Concentration using the Herfindahl-Hirschman Index (HHI), and a rule-based Mutual Fund Recommendation System were successfully implemented. These techniques enhanced the platform's ability to support investment analysis, risk assessment, and strategic decision-making.

The project highlights how data analytics can transform large volumes of financial data into meaningful business intelligence. By combining statistical analysis, financial metrics, and interactive dashboards, the platform provides investors, analysts, and financial institutions with actionable insights that improve investment evaluation and portfolio management.

Overall, the Bluestock Mutual Fund Analytics Platform demonstrates practical expertise in Python, SQL, Power BI, data visualization, financial analytics, and business intelligence. It serves as a scalable foundation for future enhancements involving machine learning, cloud deployment, real-time data integration, and predictive analytics while showcasing the effective application of data analytics in the financial services industry.

References & Technologies Used

References

The following data sources, libraries, and software tools were used during the development of the Bluestock Mutual Fund Analytics Platform.

Data Sources

- Association of Mutual Funds in India (AMFI) – Mutual Fund Industry Data
- National Stock Exchange (NSE) India – Benchmark Index Information
- Bombay Stock Exchange (BSE) – Market Index Data
- Bluestock Internship Capstone Dataset
- Publicly available Mutual Fund datasets used for educational and analytical purposes

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Software and Technologies

Programming Language

- Python 3.12.10

Data Processing & Analysis

- Pandas
- NumPy

Data Visualization

- Microsoft Power BI Desktop

Development Environment

- Jupyter Notebook
- Visual Studio Code

Version Control

- Git
- GitHub

Documentation

- Microsoft Word
- Microsoft PowerPoint

Python Libraries Used

- pandas

- numpy
- matplotlib
- seaborn
- scipy
- pathlib
- datetime

Analytical Techniques Implemented

- Exploratory Data Analysis (EDA)
- ETL (Extract, Transform, Load)
- Data Cleaning & Validation
- Feature Engineering
- Historical Value at Risk (VaR)
- Conditional Value at Risk (CVaR)
- Rolling 90-Day Sharpe Ratio
- Investor Cohort Analysis
- SIP Continuity Analysis
- Portfolio Concentration Analysis (HHI)
- Mutual Fund Recommendation System
- Interactive Dashboard Development

Project Deliverables

The following deliverables were successfully completed as part of the Bluestock Mutual Fund Analytics Platform:

- ETL Pipeline using Python
- Cleaned and Processed Datasets
- Power BI Interactive Dashboard (.pbix)
- Dashboard PDF Export
- Dashboard Page Screenshots
- Advanced Analytics Jupyter Notebook
- Risk Analysis Reports (VaR & CVaR)

- Rolling Sharpe Ratio Visualization
- Cohort Analysis Report
- SIP Continuity Report
- HHI Concentration Report
- Rule-Based Mutual Fund Recommendation System
- Final Project Report
- Project Presentation
- GitHub Repository with Complete Source Code

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I would like to express my sincere gratitude to **Bluestock Fintech** for providing the opportunity to work on this capstone project as part of the Data Analyst Internship Program. This project provided valuable hands-on experience in Data Analytics, Business Intelligence, Financial Analytics, Dashboard Development, and Data Engineering. The knowledge and practical skills gained during this internship have significantly strengthened my understanding of real-world analytics workflows and professional project development.

Thank You

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